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Editor-in-Chief  
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Dear Editor,

We would like to submit our manuscript entitled “*Adaptive Multi-Representation Fusion via Dual-Channel PCNN with Multi-Scale Convolution for Vibration Signal Fault Diagnosis*” for consideration for publication in *IEEE Transactions on Instrumentation and Measurement*.

Vibration-based fault diagnosis is essential for rotating machinery, yet existing single-representation methods capture only partial signal characteristics, and current fusion strategies lack adaptivity to dynamically regulate heterogeneous representations. To address these gaps, this work proposes a two-stage framework: an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) that fuses STFT spectrograms and GADF images via contrast-guided dynamic weighting. Furthermore, a multi-scale convolution and channel attention enhanced VGG16 (MSCA-VGG16) classifier for robust fault identification. The main contributions are:

1. **AW-DPCNN:** To overcome the single-representation limitation, an adaptive fusion mechanism is proposed that integrates heterogeneous STFT spectrograms and GADF images via contrast-guided dynamic weighting and dual-channel convolutional stimuli, adaptively regulating per-channel contributions to suppress destructive interference.
2. **MSCA-VGG16:** To fully exploit the complementary information in the fused representations, a classification network is designed with multi-scale convolution for multi-granularity fault patterns, squeeze-and-excitation channel attention for dynamic frequency-band recalibration, and a compact embedding head for enhanced discriminability.
3. Extensive experiments on the CWRU bearing dataset, including backbone and representation comparisons, ablation studies, and noise robustness evaluations, with cross-sensor (97.45%) and cross-sampling-rate (94.81%) generalization.

Results demonstrate 99.85% accuracy on the 12 kHz drive-end benchmark, outperforming seven representative backbones. Ablation confirms that adaptive weighting contributes

2.93% and multi-scale convolution adds 0.54% over their respective baselines. To contextualize our contribution within the state of the art, we conduct a dedicated comparative evaluation against representative benchmarks, particularly those from IEEE TIM. The results explicitly underscore the proposed method’s marked advantages, not only in effectively fusing multi-representational features but also in delivering robust cross-domain generalization that exceeds prior art. Noise robustness and generalization experiments further validate stable performance under realistic conditions.

We believe this work fits well within the scope of *IEEE Transactions on Instrumentation and Measurement*, particularly in intelligent signal processing, vibration signal-based fault diagnosis, and deep learning for instrumentation and measurement. This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission.

Thank you for your time and consideration.

Sincerely,

**Zonglong Bai**

On behalf of all co-authors

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